CHIH-HAO (ANDY) TSAI

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Github Portfolio: https://github.com/andytsai104/andytsai-robotics-portfolio

Education

M.S. in Robotics and Autonomous Systems — Aug. 2024 – May 2026
Arizona State University, Tempe, Arizona, United States — GPA: 3.5/4.0

B.S. in Mechanical Engineering — Sep. 2019 – Jun. 2023
National Taipei University of Technology, Taipei, Taiwan — GPA: 3.03/4.0

Professional Experience

BELIV Lab, Arizona State University — Jun. 2025 – Present
Research Assistant — Mesa, Arizona

- Built a **BC+TD3 pedestrian policy** in **CARLA** for safety-critical AV testing.
- Designed **BEV + kinematic observations** for continuous pedestrian control.
- Automated **BC training** on **ASU SOL HPC** with **Bash** job scripts.
- Increased collision rate from **0% to 45%** while reducing vehicle clearance.

Publications

- **Chih-Hao Tsai**, H. Guo, and J. Zhao, "Closed-Loop Pedestrian Policy Learning for Safety-Critical Scenario Generation in Autonomous Driving," *Modeling, Estimation and Control Conference (MECC)*, 2026 (Accepted).

Academic Projects

OmniVLA Semantic Warehouse Navigation — Jan. 2026 – Apr. 2026

- Built a **semantic navigation pipeline** with **OmniVLA-edge, ROS 2, Gazebo, and Nav2**.
- Created a **semantic goal library** linking language aliases, goal images, and navigation poses.
- Developed a **ROS 2 data collection** and **JSONL export** pipeline for VLA fine-tuning.
- Fine-tuned **OmniVLA-edge**, reaching **97.57% validation accuracy** and about **80% evaluation accuracy**.

Consensus-Based Multi-Robot Warehouse Task Allocation — Aug. 2025 – Dec. 2025

- Built **Max-Consensus** task allocation with distance-based bidding and one-task-per-robot assignment.
- Implemented a **ROS 2 + Gazebo** warehouse simulation with **2D LiDAR** robots.
- Integrated allocation results to **Nav2** goal dispatch and task-status monitoring.
- Verified **finite-time convergence** and conflict-free assignments under capacity constraints.

Pololu 3pi+ 2040 Embedded Robotics — Aug. 2025 – Dec. 2025

- Implemented **embedded autonomy** with IMU, encoders, IR sensors, bump sensors, and LCD telemetry.
- Built **line-following control** using IR calibration, lateral-error estimation, P control, and differential steering.
- Developed **ramp edge-detection** and recovery logic with sensor-triggered backup and turn maneuvers.

Vision-Based Maze Solving & Path Planning with MyCobot Pro 600 — Mar. 2025 – Apr. 2025

- Built a **ROS 2 robotic-arm pipeline** for 6-DOF maze solving from camera-captured paths.
- Created a **URDF digital twin** for motion testing in RViz and Gazebo using **SolidWorks**.
- Applied **OpenCV and A* planning** for maze preprocessing, skeletonization, and path extraction.
- Validated a **sim-to-real trajectory pipeline** on physical hardware via TCP/IP control.

Robot Kinematics Simulation — Feb. 2025 – Mar. 2025

- Built **ROS 2, Gazebo, and SolidWorks** simulation models for Dobot Magician Lite and SCARA robotic arms.
- Validated **forward/inverse kinematics** using MATLAB and Python scripts.
- Simulated **SCARA motion control** in Simulink for trajectory and controller validation.

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Bash; PyTorch, TensorFlow, OpenCV
- **Robotics & Simulation:** ROS 2, Nav2, URDF/xacro, Gazebo, RViz, CARLA, Multi-robot Systems, 3D modeling
- **Control & Embedded:** PID/PI control, kinematics, odometry/IMU integration, sensor-based control, sim-to-real validation, motion control, path planning
- **Machine Learning:** Deep Learning, Reinforcement Learning, Behavior Cloning, VLA/VLM Models, Computer Vision
- **Tools:** Linux, Git, Simulink, SolidWorks, ASU SOL HPC